

Madison A. Morgan

Class of 2027 | Aerospace Engineering at Iowa State University

PROFESSIONAL SUMMARY

Motivated Aerospace Engineering junior at Iowa State University with a focus on sustainability and a passion for advancing environmentally responsible aviation. Committed to driving innovation in flight technology while advocating for diversity and inclusivity in engineering. Planning to pursue a Master's degree in Systems Engineering. Seeking a summer internship and willing to relocate.

EDUCATION

Iowa State University, College of Engineering GPA 3.63/4.00

- Bachelor of Science in Aerospace Engineering
- Expected graduation date of 05/2027
- Sustainability Focus

Lewiston High School - Lewiston, Idaho GPA 4.00/4.00

- Class of 2023 Valedictorian
- National Honor Society Graduate

PROFESSIONAL EXPERIENCE

Open UAS: Design Team Member, 2025-current

- Utilize Design for Manufacturing (DFM) techniques to design and 3D print a configurable UAV.
- Optimize aircraft design using results from Computational Fluid Dynamics simulations in Star-CCM and XFLR5 airfoil analysis.
- Design SolidWorks 3D model of fuselage, wings, tail assembly, internal structure, etc.
- Training as Design Team Lead for 2026-2027.

ISU Department of Residence: Resident Assistant, August 2024 – current

- Lead residents' development through leadership, mentoring, and networking.
- Coordinate with directors and RAs to ensure the security and safety of residents.

Sigma Gamma Tau (SGT) Aerospace Engineering National Honor Society: President, 2025

- Rebuilt Iowa State's SGT branch membership interest by connecting SGT with the American Institute of Aeronautics and Astronautics (AIAA).

Schweitzer Engineering Laboratories: Component Development Engineering Intern, Summer 2024

- Researched industry technology to present options for compliant in-house alternative processes, and aided mechanics in implementing these alternative processes.

ISU Cyclone Rocketry: Mechanical Design Team Member, 2023-2024

- Mentored a small team in the SolidWorks Design of metal parts for the club rocket
- Complete Finite Element Analysis (FEA) of parts before machining for final construction

Schweitzer Engineering Laboratories: Sheet Metal Engineering Intern, Summer 2023

- Coded a co-bot arm inspection station, wrote SOPs, and completed 3D design projects for process improvement of electroplating and metal part punching.

WiSE: Women in Science and Engineering (WiSE) at Iowa State Member, 2023-Current

- Lead programs for the WiSE Community, fostering academic retention.

CONTACT INFO

Lewiston, Idaho

(208) 305 1969

mamorgan@iastate.edu

LinkedIn as Madison A. Morgan



SKILL SETS

INDUSTRY KNOWLEDGE

- Manufacturing: SolidWorks, CAD, DFM, 3D Printers/Slicers
- Aircraft Analysis: Star-CCM, XFLR5, FEA, CFD
- Technical Communication and Writing, GitHub

CODING PROFICIENCIES

- Linux/bash
- C
- Python
- MATLAB
- Programmable Logic Controllers

INTERPERSONAL

- Team Leadership and Coordination
- Public Speaking
- Mentoring and Networking

ACCOMPLISHMENTS

Iowa State Dean's List

2024 Resident Assistant of the Year - Helser Hall

2024-Current: Sigma Gamma Tau - Iowa State Aerospace Honors Society

TECHNICAL PROJECTS

UAV Design: 2025-Present, Drag Profile Analysis and AIAA Conference Paper

LUNAR ORBITER MISSION: 2025, Orbital Mechanics and Research Paper

TECHNICAL INSTRUCTION WRITING: 2025, Construction and Concept Descriptions

AIRCRAFT CONCEPTUAL DESIGN: 2025, AirHeart International Plane Design

SOLIDWORKS AIRCRAFT REPLICA: 2024, Airbus Beluga Assembly

EMISSIONS IN AEROSPACE REPORT: 2024, Technical Research Paper

PYTHON ORBITAL MECHANICS: 2023, Rocket Launch Physics Code Animation

AIRSHIP RESEARCH PROJECTS: 2021–2023, Airship Efficiency and Lighter-Than-Air Craft